

MICROPROCESOARE SI LIMBAJE DE ASAMBLARE

Calculatoare 3 2025-2026

Curs 1 Sapt 1 30 septembrie 2024 14:00-16:00

serban@upit.ro

Modul de notare:

Examen:	50%	
Laborator:	30%	A2
Test (Midterm):	20%	A1

Planificarea activitatilor:

Laborator:

- saptamanal, conform orarului;
- sapt. 13, 14 refaceri laboratoare;

Curs:

- saptamanal, conform orarului;

Test (Midterm): in saptamana a 12-a (15 decembrie - 20 decembrie)

OBS. Este nevoie de nota minima 5 la toate activitatile (semestru – A1, A2 si sesiune – Examen!!! pentru promovare)

Continut

Introducere

Evolutia microprocesoarelor

Criterii de clasificare ale microprocesoarelor

Structura microprocesoarelor

Mecanisme specifice Microprocesoarelor

Microprocesoare pe 8 biti

Familia de microprocesoare x86

Familia de procesoare ARM

Bibliografie

The Design of a Microprocessor – Wilhelm G. Spruth – Springer

Microprocessor Design – Grant McFarland – McGraw-Hill – 2006

Microprocessor Interface Design – J.D. Nicoud – Chapman & Hall

Microprocessors and Microcomputers Hardware and Software - Ronald J. Tocci – Pearson

Intel Microprocessors – Barry Brey

Z-80 Microprocessor Architecture, Interfacing, Programming and Design - Ramesh Gaonkar

Introduction to 80X86 Assembly Language and Computer Architecture - Richard C. Detmer - Jones & Bartlett Pub (2001)

X86 Assembly Language and C Fundamentals – Joseph Cavanagh – CRC Press

ARM Assembly Language Fundamentals and Techniques, Second Edition - William Hohl

Fundamentals of Embedded Software Where C and Assembly Meet – Daniel Wesley Lewis

INTRODUCERE

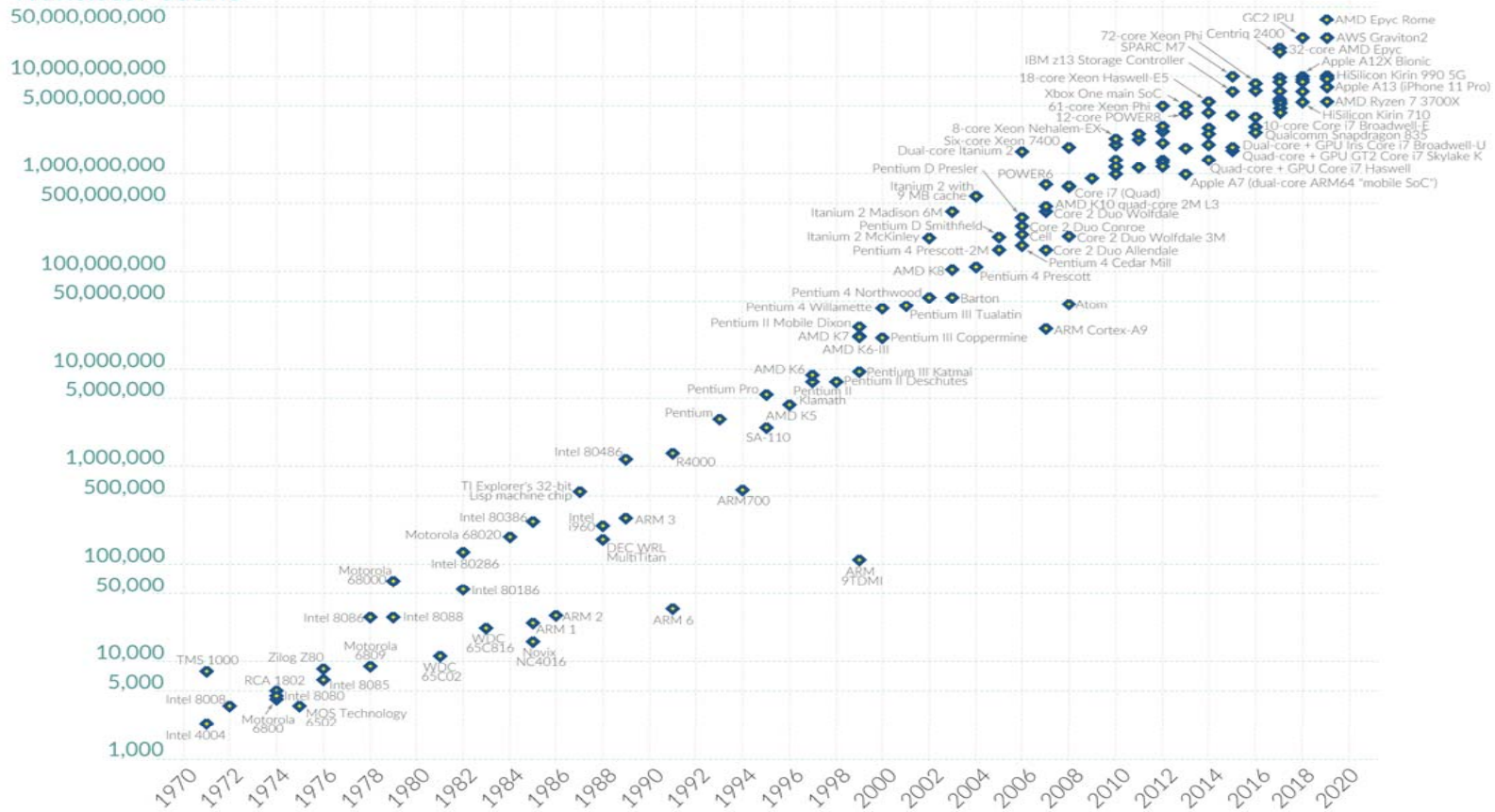
Moore's Law: The number of transistors on microchips doubles every two years

Moore's law describes the empirical regularity that the number of transistors on integrated circuits doubles approximately every two years.

This advancement is important for other aspects of technological progress in computing – such as processing speed or the price of computers.

Our World
in Data

Transistor count



Data source: Wikipedia (wikipedia.org/wiki/Transistor_count)

OurWorldinData.org – Research and data to make progress against the world's largest problems.

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Evolutia tehnologica la fabricatia microprocesoarelor

10 μm – 1971	INTEL Ted Hoff
6 μm – 1974	
3 μm – 1977	
1.5 μm – 1981	
1 μm – 1984	
800 nm – 1987	
600 nm – 1990	
350 nm – 1993	
250 nm – 1996	
180 nm – 1999	
130 nm – 2001	
90 nm – 2003	
65 nm – 2005	
45 nm – 2007	
32 nm – 2009	
22 nm – 2012	
14 nm – 2014	
10 nm – 2016	
7 nm – 2018	
5 nm – 2020	
3 nm – ~2022	

Future
[2 nm](#) – 2024

Evolutia istorica microprocesoare

IBM 360 – Consola (Video + tastatura)

Busicom – Japan

Tedd Hoff

1972 – 4004 (4b)

1973 – 8008 (8b)

1974 – 8080 (8b)

Zilog – Z80 (8b) (1975)

1976 – 8086 (16b)

1982 – 80286 (16b)

AMD

PC XT (MCU+DRAM+HDD)

1985 – 80386 (32b)

1988 – 80486 (32b)

PC AT

Pentium

Pentium 2

Pentium 3